



Single-board computer with ARM processor

VGA, touchscreen and TFT display." Other requirements include analogue inputs/outputs (I/Os), digital I/Os, voltage and current input.

Sathya shares, "Longevity of up to three-five years and heavy technical support have always been a concern for the designers. They also look for embedded operating-system support and possible security issues."

SBCs can be used in a variety of applications and the processor requirement varies. "For example, in an industrial environment, where the importance is given to ruggedness of the box rather than the CPU performance, the preference would be fanless processors like those from Intel (Atom) and VIA. But for billing systems, which call for more processing power, the SBC may require a dual-core or higher-performance processor," explains Sathya. "For a network appliance, it will be the external interface like LAN port that plays an important role, while in a PDA or other handheld, where integration on the chip is important in compact space, it is based on ARM. For high-performance defence applications, where the processing speed (in

million instructions per second (MIPS)) plays a key role in the CPU selection, it will require PowerPC core," he adds.

Manikandan shares, "Industrial computers focus on improved mechanics to overcome the limitations posed by the standard consumer PC set-up." The ultimate deciding factors thus are size, power consumption, availability of compatible software and peripheral hardware, and the low mean time between failures (MTBF), all at the most affordable price point.

ARM-core SBCs

Single-board computers with ARM-based microprocessor have made a big impact on the embedded sector. Though the first truly functional SBC was based on Intel's C8080A and C1702A (Intel's first EPROM), today, there are hundreds of processors that are used in single-board computers. ARM, which was prevalent in the handset space, has entered single-board computers with its Cortex A series processors.

At the *Embedded Systems Conference Silicon Valley*, a single-board computer based on 800MHz Cortex A8 CPU was

showcased by Microsys. An interesting aspect of this single-board computer is its cost, which is roughly \$100 to \$300 less than an Intel-based board. Freescale's single-chip solution as opposed to two-chip solution for Intel-based board is one of the contributors to the price drop.

Rajesh shares, "Some of the compact, full-featured SBCs based on Cirrus EP9302 ARM9 CPU provide a standard set of on-board peripherals. These low-cost Linux computers are completely manufactured in India to benefit the designer's local needs."

"The Mini2440/MICRO2440 is a low-cost single-board computer based on Samsung's S3C2440 ARM920T core. It supports WinCE 5.0/6.0, Linux, Android and Ubuntu, making it simple and suitable for applications like industrial control, check-out facilities, meters, safe monitoring, medical instruments, high-end embedded applications and intelligent terminals," shares Prithesh Patel, managing director, Amp Tronics.

Beagle Board is another popular single-board computer that is powered by a 1GHz ARM Cortex A8 processor.

The ITX world

Many vendors today offer VIA Technologies' latest SBCs based on the new super-scalar Nano-E processors. "The EPIA embedded boards from VIA Technologies features low-power embedded processors, core logic, networking, connectivity and multimedia components. Vendors develop their own SBCs based on a variety of form factors from the company, namely, Mini-ITX, Nano-ITX, Pico-ITX and Pico-ITXe embedded boards" shares Sathya.

Mobile-ITX is currently the smallest form-factor (6x6cm²) board based on x86 platform. It brings ultra-compact and extremely light-weight devices to the market. The Pico-ITXe form factor (10x7.2cm²) was adjudged as 'Best Choice' at *Computex Taipei 2009* for its small form factor and stackability. Pico-ITXe is adopted by the Small Form